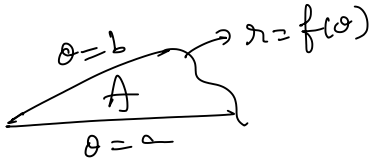
Lecture 18

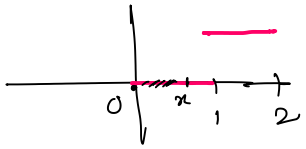
Friday, 18 September 2026 8:22 AM

Recap:



- Area of the region $A = \frac{1}{2}\int_{\theta=a}^{b} (f(\theta))^2 \, d\theta$

Eg1: $f(x) = [x] \quad 0 \le x \le 2$.



$$A(x) = \int_0^x f(t)\,dt$$

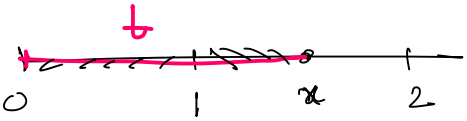
$$A(x) = \int_0^x [t]\,dt \qquad 0 \le x < 1$$



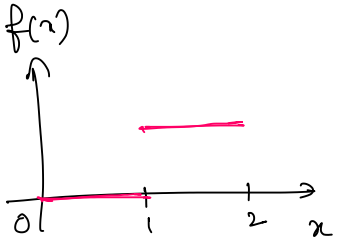
$[t] = 0$

$$A(x) = \int_0^x 0 \, dt = 0 \qquad 0 \le x < 1$$

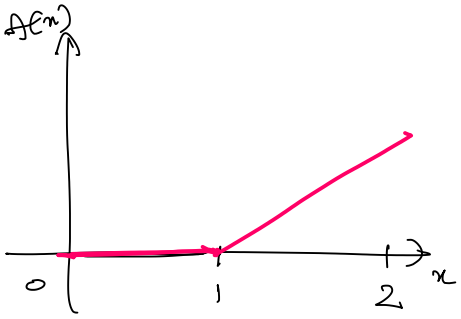
$$1 \le x < 2$$



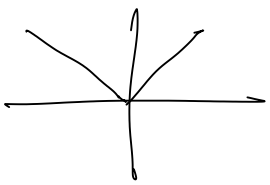
$$A(x) = \int_0^x f(t)\,dt = \int_0^1 0 \, dt + \int_1^x 1 \, dt = x - 1 \qquad 1 \le x < 2$$



$\int f(x)\,dx$



Eg2: $f(x) = |x| \qquad -1 \le x \le 1$

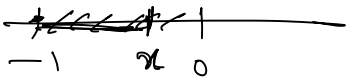


$$A(x) = \int_{-1}^{x} f(t)\,dt$$

$$f(x) = \begin{cases} -x & -1 \le x < 0 \\ x & 0 \le x < 1 \end{cases}$$

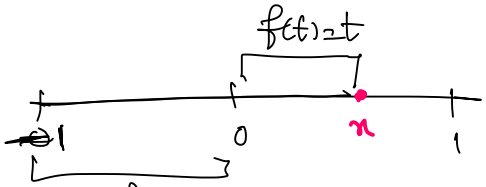
Condition $-1 \le x < 0$:

$$A(x) = \int_{-1}^{x} -t\,dt$$



$$= \frac{1-x^2}{2} \qquad -1 \le x < 0$$

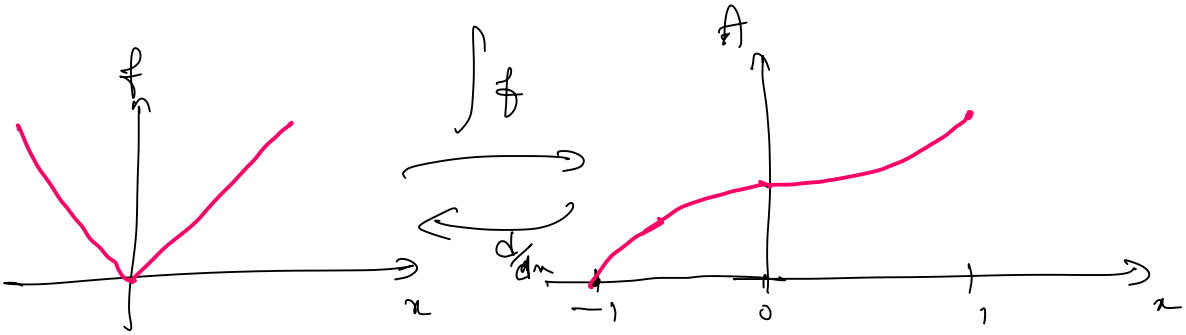
Consider: $0 \le x < 1$



$$A(x) = \int_{-1}^{x} f(t)\,dt = \int_{-1}^{0} -t\,dt + \int_{0}^{x} t\,dt$$

$$= \frac{1}{2} + \frac{x^2}{2} = \frac{1+x^2}{2} \qquad 0 \le x < 1$$

$$A(x) = \begin{cases} \frac{1-x^2}{2} & -1 \le x < 0 \\ \frac{1+x^2}{2} & 0 \le x < 1 \end{cases}$$



Q: What's the connection b/w Diff & Integration?

Fundamental Theorem of Calculus (FTC)

If f is continuous at $x = p$

then $A(x) = \int_a^x f(t)\,dt$ is diff'ble at $x = p$.

Further $A'(p) = f(p)$.

How to evalute $\int_a^b f(t)\,dt = ?$

Primitive: If $P' = f$ then P is called primitive of f.

Eg: (1) $(x)' = 1$ $\quad P(x) = x,\ f(x) = 1$

Notation: $P(x) = \int f(x)\,dx$

(2) $\left(\frac{x^2}{2}\right)' = x \qquad \int x\,dx = \frac{x^2}{2}$

(3) $(\sin x)' = \cos x \qquad \int \cos x\,dx = \sin x$

Second FTC:

If P is a primitive of the function f

then $\int_a^x f(t)\,dt = P(x) - P(a)$.

Eg: If $f'(x^2) = x$, $x > 0$, $f(1) = 1$

then find $f(x) = ?$

Sol: let $y = x^2$

$f'(y) = \sqrt{y}$, $y > 0$

$f(1) = 1$,

$$\int_{t=1}^{y} f'(t)\,dt \overset{SFTC}{=} f(y) - f(1)$$

$$\Rightarrow \int_1^y \sqrt{t}\,dt = f(y) - 1$$

$$\int_a^x f'(t)\,dt = f(x) - f(a)$$

$$\Rightarrow \frac{2}{3} t^{3/2} \Big|_{t=1}^{y} = f(y) - 1$$

$$\Rightarrow \frac{2}{3} \left[y^{3/2} - 1 \right] = f(y) - 1$$

$$\Rightarrow f(y) = 1 + \frac{2}{3} \left[y^{3/2} - 1 \right].$$

Eg 2: $f'(\sin^2 x) = \cos^2 x, \quad f(1) = 1, \quad x > 0$

Find $f(x) = ?$

$$\int_a^x f'(t)\,dt = f(x) - f(a)$$